Fregata Flying Liferaft

Empennage Design and Static Stability Analysis

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Abstract

The goal of the Fregata project is to design a fully autonomous Unmanned Aerial Vehicle (UAV) for maritime search and rescue purposes. The fixed-wing drone is also designed for compact storage and deployment from any water vessels, particularly coast guard ships. This project explored many possibilities of designing a fixed-wing drone, implementing complex control systems to minimise human error. This report focuses on the methodology adopted for empennage design and static stability of the UAV. The aircraft was found to be stable with the results obtained from static stability analysis, tail and primary control surfaces sizing.

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1 Introduction

The goal of Fregata is to search for victims at sea, descend to targeted area, deploy liferaft while sending signal back to base and cruise back to the nearest vessels. Here is a brief summary of Fregata's specifications: our UAV aims to deliver payload weighed up to 10 kg, with range 70 km, cruising at a velocity of 58.3 ms^{-1} at an altitude of 500 m.

Flight stability, performance and controls depends heavily on baseline configurations and decisions made by the Aerodynamics, Structures and Propulsion and Systems teams. Static stability analysis ensures the aircraft is in equilibrium in cruising flight, which was done simultaneously with the design of tailplane and primary control surfaces.

1.1 Review of Similar Aircraft

One fixed-wing Unmanned Aerial Vehicle (UAV) that is very similar to our UAV, Fregata, is the $Wingcopter$ 198. It is a delivery drone with modular payload, speed and range that is very close to Fregata's specifications. [1] $Wingcopter$ 198 utilises a V-tail which has several advantages and disadvantages compared to Fregata. V-tail has a smaller wetted area compared to any other tail configuration, results in lower structural weight. Nonetheless, a longer rear fuselage is needed to counter 'snaking' or yawing.

The other example is the DRS RQ-15 $Neptune$. The implementation of H-tail with pusher configuration together with the similar specifications, inspired the empennage design of Fregata. [2] Other UAVs that implement H-tail are the $Plyrotech\ XV-H$ that has search and rescue purposes [3], and the $ElevonX\ Tango$ that has vertical take-off and landing (VTOL) capabilities. [4].

1.2 Methodology

The empennage plays a vital role in providing longitudinal and directional stability in flight. The details and calculations for each step were explained in the following sections. The iterative design process for both horizontal and vertical tailplanes was simplified below:

1. Identify the tail design requirements such as trim, stability, structural constraint: Compact storage is our largest constraint in this case as it needs to be able to store and take off from coast guard ships.
2. Select tail configuration: H-tail configuration was chosen to reduce structural weight of the overall empennage. The vertical stabilisers were split into two with smaller span, thus reducing storage space (similar to the aircraft $Lockheed\ Constellation$). The two fins were able to provide more control at low speeds as well, which is the range of speed our UAV will be flying at.
3. Select horizontal tail location.
4. Select tail volume coefficient.
5. Determine optimum tail arm.
6. Determine tail planform area.
7. Determine airfoil selection.
8. Determine aspect and taper ratios, sweep and dihedral angles.
9. Determine setting angle.
10. Calculate span, mean aerodynamic centre, root and tip chord.

11. Analyse longitudinal and directional stability and optimise.

The whole design process was done in close collaboration with the Aerodynamics team where they determined the airfoil, aspect and taper ratios, sweep and dihedral angles of both tailplanes to account for prop-wash and end-plate effect.

Some important parameters from the Aerodynamics and Propulsion and Systems Team used in the calculations were summarised in Table 4 in the Appendix. The wing was placed at the centre of gravity so that it does not generate any pitching moment. The constraining pylon length mounted on the wing was also taken into consideration when placing the wing to make sure there is sufficient space for the tailplane.

2 Trim Analysis

The performance of the UAV at trimmed flight is important as the vehicle will spend about 30 minutes cruising, which is half of the flight time for our primary mission profile. In this section, the methodology used in trim analysis will be explained including the equations needed. Values relating to the empennage will only be substituted in after the geometry parameters are finalised.

First of all, the lift required by the tailplane was found using the relations below for sizing.

$$C_{L_{H_{trim}}} = \frac{C_{L_{trim}} - C_{L_{W_{trim}}}}{\eta_H \frac{S_H}{S_{ref}}} \tag{1}$$

where

$$C_{L_{trim}} = \frac{W_{avg}}{0.5\rho V_{cruise}^2 S_{ref}} = 0.3424 \tag{2}$$

given that the wing lift coefficient at cruise is 0.3467. η_H is the tail efficiency factor which was taken to be 1 throughout the analysis, $\rho = 1.225 kgm^{-3}$ when the fixed-wing drone is cruising at an altitude of 500 m. Desired cruise speed, v_{cruise} is 58.3 ms^{-1}.

The contributing pitching moments were calculated using the relations below to determine the elevator deflection required at trimmed flight.

$$C_{M_0} = C_{M_{0_{wf}}} + \eta_H \bar{V}_H a_h \frac{S_H}{S_{ref}}(i_H - \epsilon_0 + (1 - \frac{d\epsilon}{d\alpha})\alpha_0) \tag{3}$$

where

$$C_{M_{0_{wf}}} = C_{M_{af}}(\frac{AR}{AR + 2}) + 0.01\alpha_{twist} \tag{4}$$

$C_{M_{af}}$ is the wing airfoil section pitching moment which can be found from wind tunnel testing. It is assumed to be -0.01 in this case as our drone is travelling at a much lower speed.[5] The wing aspect ratio is 6.83, referring to Table 4.

The total pitching moment was found by summing all the contributing pitching moments.

$$C_M = C_{M_0} + C_{M_\alpha}\alpha + C_{M_{iH}} i_H + C_{M_{\delta_e}}\delta_e \tag{5}$$

C_{M_α} was explained under Section 3.3, using Equation 24. $C_{M_{i_H}}$ was taken to be three times $C_{M_{\delta_e}}$ as suggested by [6].The elevator effectiveness was obtained using Equation 14.

The summation of lift from all components gives the aircraft total lift coefficient.

$$C_L = C_{L_w} + C_{L_\alpha}\alpha + C_{L_{iH}} i_H + C_{L_{\delta_e}}\delta_e \tag{6}$$

where

$$C_{L_\alpha} = a_v = a_w + a_H \eta_H \frac{S_H}{S_{ref}}(1 - \frac{d\epsilon}{d\alpha}) \tag{7}$$

is the lift curve slope of the whole aircraft in rad^{-1}.

3 Longitudinal Stability

3.1 Horizontal Tailplane Sizing

The main purpose of the horizontal tailplane is to ensure zero pitching moment at trimmed flight and to allow the aircraft to function under the maximum load case of 4.5 obtained from the V-n diagram.

An adjustable tail was chosen over a fixed tail due to higher maneuverability with larger tail setting angle. It was preferred over an all moving tail due to safety - to ensure there is some control in the case of a failure in the tailplane mechanism.

Two options to mount the horizontal tailplane were considered: on the fuselage and using a twin boom. Initially, the twin-boom H-tail was considered to accommodate a pusher propeller. The longer tail arm also allows the tailplane to give more control to the aircraft. However, given that one of our design objectives is to be compact, the aircraft size is restrained to 4.5 m by 1.5 m to be able to take off from the deck of coast guard ships. Therefore, this idea was disregarded given the storage limitations. Finally, the tail configuration proceeded with H-tail mounted on fuselage.

The iterative design process starts with the tailplane location on the fuselage. To maintain stability, the tailplane was placed as far back as possible, only constrained by the pusher propeller. The aerodynamic centre of the tailplane, x_{ac_H}, was placed at 1.20 m from the nose, with a fuselage length of 1.35 m. A few centimeters were spared for the attachment of the tailplane to the fuselage. The optimised tail arm was then calculated using Equation 8 by choosing an arbitrary tail volume coefficient, $\bar{V}_H$.

$$l_{opt} = K_c\sqrt{\frac{4\bar{C}S\bar{V}_H}{\pi D_f}} \tag{8}$$

where K_c is the correction factor to account for the error in the assumptions made in calculation and D_f is the width of the fuselage. $K_c = 1$ was used in this case because the aft portion of the fuselage has a conical shape as suggested by [7].

The $\bar{V}_H$ was then adjusted iteratively to ensure that x_{ac_H} is at 1.20 m using l_{opt} obtained above, producing a final value of 0.36. Next, the tailplane area was obtained using Equation 9.

$$S_H = \frac{\bar{V}_H \bar{c} S_{ref}}{l_{opt}} \tag{9}$$

The maximum span of the horizontal tailplane was restricted by the back VTOL propellers and minimum span is restricted by the diameter of the pusher propeller blades. The aspect ratio was decided by [8] to keep it out of the prop-wash, maximising aerodynamics efficiency and to match the optimised tailplane area. The span and aeerodynamic chord of the tailplane was calculated using empirical equations 10 and 11.

$$b_H = \sqrt{S_H AR_H} \tag{10}$$

$$c_H = \frac{S_H}{b_H} \tag{11}$$

The upper and lower bound of the vertical position of the tailplane is given by:

$$\begin{aligned} h &> l_{opt} \cdot tan(alpha_{stall} - i_w + 3) \\ h &< l_{opt} \cdot tan(alpha_{stall} - i_w - 3) \end{aligned} \tag{12}$$

The wing wake expands as it travels downstream and it was accounted for in Equation 12. This is to ensure that the tailplane is out of the wing wake region during stall. The lower bound was found to be 0.151 m. Hence, the tailplane is placed at the centre of gravity axis which is the maximum radius of the fuselage, 0.177 m, to give the aircraft adequate stall recovery ability without adding pitching moment to the overall calculation.

The tail setting angle was initially set at -1 ° but it was later increased to 0 ° as suggested by the Simulations team as the tailplane was very responsive and was destabilising the aircraft. Once the initial sizing of the horizontal tailplane was done, longitudinal stability of the drone was analysed and the design was optimised.

3.2 Elevator Sizing

Elevator is used to improve the performance of the horizontal tailplane by adding camber to the tailplane, providing additional lift when deflected.

The span ratio of the elevator was initially set at 1 to maximise its efficiency. However, it was changed and finalised to be 0.9 to fit a servo actuator as requested by the Propulsion team [9] for a minimum space of 5 cm. Since the horizontal tailplane is tapered, two elevators will be needed.

The elevator effectiveness was calculated using Equations 13 and 14, which is a measure of how effective the elevator deflection in producing the desired pitching. It is primarily determined by the geometry of the horizontal tail.

$$C_{L_{\delta_e}} = a_H \eta_H \frac{S_H}{S_{ref}} \frac{b_e}{b_H} \tau_e \tag{13}$$

$$C_{M_{\delta_e}} = a_H \eta_H \bar{V_H} \frac{b_e}{b_H} \tau_e \tag{14}$$

The maximum elevator deflections was calculated using Equation 15.

$$\delta_e = -\frac{f_1 a_v + C_{M_\alpha}(C_L - C_{L_w})}{C_{M_{\delta_e}} a_v - C_{M_\alpha} C_{L_{\delta_e}}} \tag{15}$$

where

$$f_1 = \frac{T z_T}{0.5 \rho v_{cruise}^2 S_{ref}} + C_{M_0} \tag{16}$$

The thrust produced by the pusher propeller at cruise is 109.91 N. Zero lift pitching moment, C_{M_0} was found using Equation 3. The thrust line relative to the centre of gravity axis, z_T, varies in different cases and is shown in Table 1. This occurred with a shift of load in the fuselage and also a change in CG position in different cases.
Based on the different cases presented in Figure 1, the maximum deflections of the elevator was then decided to be +10/-15 °.
The summary of the design parameters of the horizontal tailplane and elevator is shown below.

Table 1: Thrust Line Positions in Three Cases

With payload	Without payload	Zero fuel, without payload
-0.0355 m	-0.1403 m	0.0355 m

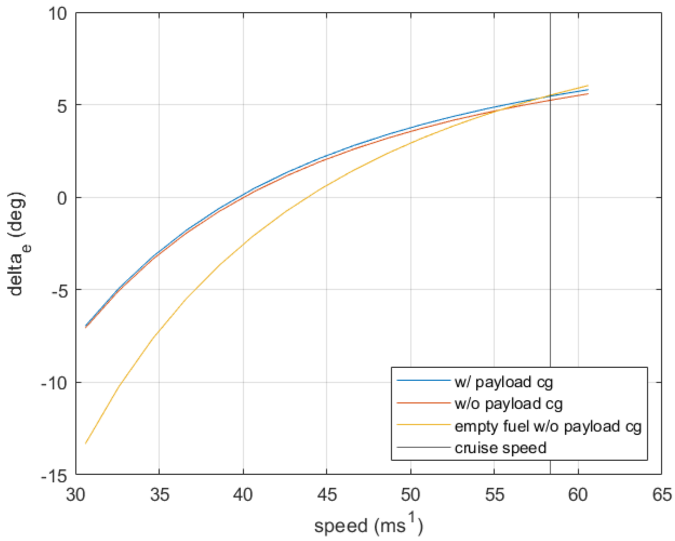

Figure 1: Elevator deflections with a variation of speeds

Table 2: Horizontal Tailplane and Elevator Parameter

Parameters	Horizontal Tailplane	Elevator
Airfoil	NACA 642-015	
Lift curve slope, rad^{-1}	3.352	
Tail volume coefficient, $\bar{V}_H$	0.36	
Area, m^2	0.1548	0.0418
Span, m	0.7421	0.6679
Root chord, m	0.2811	0.0843
Tip chord, m	0.1687	0.0506
Taper ratio, λ	0.6	0.6
Effectiveness, τ		0.5
x-position, x_{acH}	1.20	
z-position, z_{acH}	0	
Tail arm, m	0.5567	
Tail setting angle, deg	0	
Zero lift angle of attack, α_{0_H}	-2.71	
Deflections, deg		+10/-15

The z-positions are relative to the centre of gravity axis. Trim tab is not required since the elevator is sufficient to provide the necessary lift to keep the aircraft at trim.

3.3 Stick Fixed Analysis

Assuming clockwise pitching moment to be negative, the condition for the aircraft to be stable is $\frac{dC_M}{d\alpha} < 0$. The aircraft should be able to react with negative pitching moment when the angle of attack increases. The neutral point, x_{np}, the point where the aircraft is neutrally stable in pitch, needs to be after the centre of gravity, x_{cg}. It was suggested for the static margin, K_n to stay within the range of 0.1 - 0.7 for the vehicle to be controllable. The larger the magnitude of K_n, the larger the moment produced by the aircraft to counter the pitching moment but it is more sluggish. Conversely, a smaller magnitude means the vehicle is more responsive but also more unstable.

The effect of downwash from the wing to the tailplane has significant effect on the performance. The relationship between the downwash and angle of attack is

$$\frac{d\epsilon}{d\alpha} = \frac{2a_w}{\pi AR} \tag{17}$$

The effective angle of attack as a result of wing downwash was calculated to obtain the final designed lift coefficient.

$$C_{L_H} = a_H(\alpha_H + \tau_e \delta_e) \tag{18}$$

where

$$\alpha_H = \alpha_w + i_H - \epsilon = -2.8020 \tag{19}$$

$$\alpha_w = \frac{C_{L_w}}{a_w} + \alpha_0 = -2.705 \tag{20}$$

$$\epsilon = \epsilon_0 + \frac{d\epsilon}{d\alpha}\alpha_0 = 0.0967 \tag{21}$$

$$\epsilon_0 = \frac{2C_{L_w}}{\pi AR} = 0.1174 \tag{22}$$

The lift coefficient obtained at maximum elevator deflection of 5 ° at trim is -0.024. Using the initial design values, the whole aircraft lift curve slope is 5.1207 rad^{-1}. The change of whole aircraft pitching moment and lift coefficients against the angle of attack with different elevator deflections is presented below. The trim plot shows that the pitching moment at zero angle of attack with 5 ° elevator deflection is zero, producing a lift of 0.3424. The results are in agreement with values obtained in Section 2.

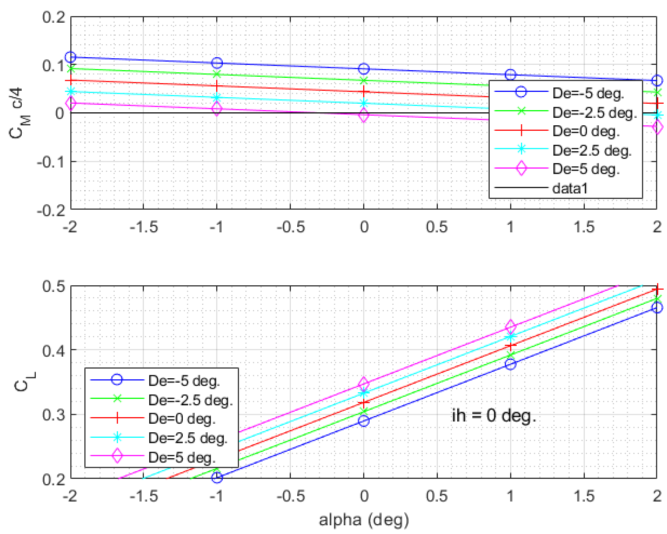

Figure 2: Trim Plot

As mentioned above, $\frac{dC_M}{d\alpha}$ needs to be negative for the aircraft to be pitch stable. For the aircraft to be speed stable, $\frac{dC_M}{dC_L} < 0$. [10] Stick fixed analysis was done to determine the trimmability of our

UAV since it is fully autonomous. The elevator contribution to C_{L_H} is constant at cruise speed. $\frac{dC_M}{dC_L}$ is given by:

$$\frac{dC_M}{dC_L} = C_{M_{CL}} = (\bar{x}_{cg} - \bar{x}_{ac_w}) - \bar{V_H}\frac{a_H}{a_v}(1 - \frac{d\epsilon}{d\alpha}) = -0.1322 \tag{23}$$

and multiplying 23 by the whole aircraft lift curve slope gives

$$\frac{dC_M}{d\alpha} = C_{M_\alpha} = \frac{dC_M}{dC_L}a_v = -0.6770 \tag{24}$$

Therefore, the aircraft is pitch and speed stable.

The aircraft's neutral point is determined when $\frac{dC_M}{dC_L} = 0$, solving for the aftmost CG position.

$$x_{np} = x_{ac_w} + \bar{V_H}\frac{a_H}{a_v}(1 - \frac{d\epsilon}{d\alpha}) = 0.6589 \tag{25}$$

The measure of static stability, known as the static margin, can then be found easily, which is expressed as the distance between the neutral point and the centre of gravity.

$$K_n = -\frac{dC_M}{dC_L} = \frac{x_{np} - x_{cg}}{\bar{c}} = 0.1322 \tag{26}$$

By definition, $K_n > 0$ for a stable aircraft.

To decide on the size of actuators for the elevator, the hinge moment was computed using Equation 27 with maximum elevator deflection of 15 °.

$$H = C_H(\frac{1}{2}\rho v_{cruise}^2 S_e c_e) = 2.8655Nm \tag{27}$$

where

$$C_H = C_{L_H} l_{opt} + C_{M_{\delta_e}} \delta_{e_{max}} \tag{28}$$

4 Directional Stability

4.1 Vertical Tailplane Sizing

Several assumptions were made to simplify the stability analysis process when sizing the vertical stabiliser of our aircraft:

1. It is assumed that the rolling moment produced by both fins are equal and opposite so they cancel each other out.
2. Both fins are assumed to be identical and assert the same force. Hence, the analysis is done based on conventional tail design and divide by two.

Removal of these assumptions will significantly increases the complexity of the analysis, and that is not required at this stage.

Adjustable fins were chosen to ensure stability in extreme cases such as crosswind at cruise and during a turn. All moving vertical stabilisers were not considered here as our aircraft will be operating in subsonic regions throughout the flight. Vertical tailplanes are used to counter any yawing moment in these three cases: crosswind, turn and adverse yaw. The analysis for extreme cases is presented below.

The first extreme case was to consider the crosswind at cruising flight. The Mediterranean tropical storm has speed ranged 30-53 knots which is approximately 11-27 ms^{-1}. Hence, by using a maximum wind speed of 27.7 ms^-1 acting perpendicularly to our aircraft in one direction, it gives a sideslip angle, β, of 25.414 $^\circ$ and wind force valued at 213.161 N calculated using Equation 29. The $\frac{S_{wet}}{S_{ref}}$ was assumed to be 8 based on initial sizing of our UAV. The side force drag coefficient, C_{D_y} was taken to be 0.08, according to typical values of 0.5-0.8 for conventional aircraft which has a fuselage length of about a ten times higher than our aircraft. [7]

$$F_{wind} = \frac{1}{2}\rho v_{wind}^2 S_{wet} C_{D_y} \tag{29}$$

$$\beta = tan^{-1}(\frac{v_{wind}}{v_{cruise}}) \tag{30}$$

The largest sideslip angle for the drone to keep in control is 30 $^\circ$.

The aircraft relative speed under maximum wind speed condition was found using Equation 31 to get the dynamic pressure at that velocity. The relative speed is the summation of the wind speed and the aircraft cruise speed. The ratio of dynamic pressures was then included in the rudder derivatives presented in Equation 35.

$$v_t = \sqrt{v_{wind}^2 + v_{cruise}^2} \tag{31}$$

The yawing moment coefficient $C_{N_{wind}}$, required to counter the crosswind is 0.0656.

$$C_{N_{wind}} = \frac{F_{wind} \cdot l_{opt}}{0.5\rho v_t^2 S_{ref}} \tag{32}$$

In our second case which is during a turn, the maximum load factor, n_{max} is 3, taken from the V-n diagram. In this case, it was assumed that the aircraft was changing its direction only, which meant there was no rolling moment included. The lift and yawing moment coefficient was then found to be 1516 N and 0.4665 respectively. As the yawing moment required during a turn was much higher than the yawing moment required in maximum wind speed, this was considered as the largest constraint to size the fins.

$$C_{N_{turn}} = \frac{3 \cdot MTOW \cdot 9.81 l_{opt}}{0.5\rho v_t^2 S_{ref}} \tag{33}$$

Based on the analysis done by [11], adverse yaw produced by ailerons in roll is negligible hence it is not included in this analysis.

4.2 Rudder Sizing

The initial chord ratio for the rudder was set to 0.3, and the rudder effectiveness, $\tau_r = 0.5$ was obtained from Figure 5 in the Appendix. [7] This was the typical values used to size rudders and was later revised due to insufficient control in extreme cases.

The maximum rudder deflections were determined by plotting the yawing moment produced with varying deflections, after getting the required counter yawing moment in critical conditions. Note that the rudder deflection presented in the plot represents both rudders. Therefore, the maximum rudder deflection for each rudder is $\frac{34}{2} = 17^\circ$. The highest possible rudder deflection to maintain control authority is 25 $^\circ$ as suggested by [12].
The summary of the each of the vertical stabiliser is shown in Table 3.

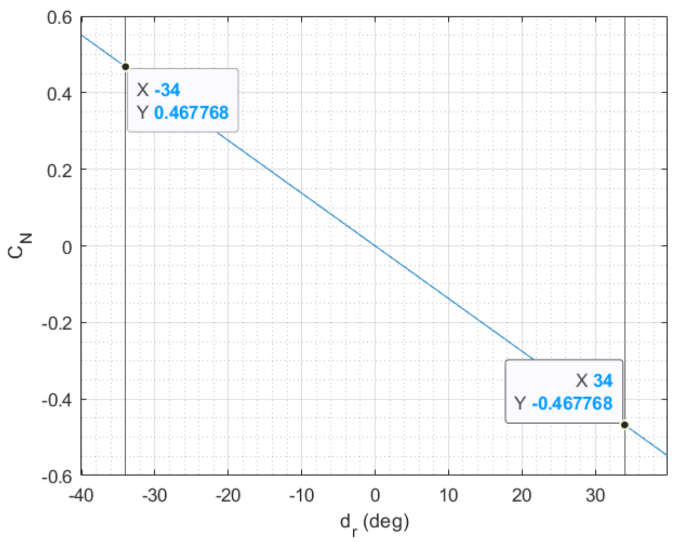

Figure 3: Yawing Moment Produced with Varying Rudder Deflection

Table 3: Vertical Tailplane and Rudder Parameter

Parameters	Vertical Tailplane (each)	Rudder (each)
Airfoil	NACA 0010	
Lift curve slope, rad^{-1}	3.003	
Tail volume coefficient, $\bar{V}_V$	0.21	
Area, m^2	0.0452	0.0154
Span, m	0.2366	0.2011
Root chord, m	0.1687	0.0675
Tip chord, m	0.1012	0.0405
Taper ratio, λ	0.6	0.6
Effectiveness, τ		0.6
x-position, x_{ac_V}	1.2	
z-position, z_{ac_V}	0	
Tail arm, m	0.5567	
Tail setting angle, deg	0	
Deflections, deg		+17/-17

Similar to the horizontal tailplane, trim tab is not necessary for the vertical stabilisers. Comparing the area of the vertical stabiliser to the horizontal stabiliser,$S_H = 0.1548m^2$, it justified the choice of not implementing a T-tail as mentioned by [8].

4.3 Yaw Stability

The control derivatives are functions of the tailplane geometry. The vertical tail volume coefficient, $\bar{V}_V$ was arbitrarily chosen to be 0.3 to start the design process. The yawing moment produced by the aircraft to counter the sideslip angle is given by:

$$\frac{dC_N}{d\beta} = C_{N_\beta} = \bar{V}_V a_V = 0.022 \tag{34}$$

This was to ensure the aircraft stay in the desired flight path.

The positive value means that when a positive sideslip angle is present, the aircraft will produce a positive yawing moment (anti-clockwise) to counter the sideslip.

The second derivative calculated to ensure yaw stability is the change of yawing moment with rudder deflections, shown below.

$$\frac{dC_N}{dr} = C_{N_{\delta_r}} = -a_v \bar{V}_V \frac{q_{v_t}}{q} \tau_r \frac{b_r}{b_v} = -0.0138 \tag{35}$$

The rudder derivative is plotted against a range of tail volume coefficient in Figure 4 to get the optimised $\bar{V}_V$.

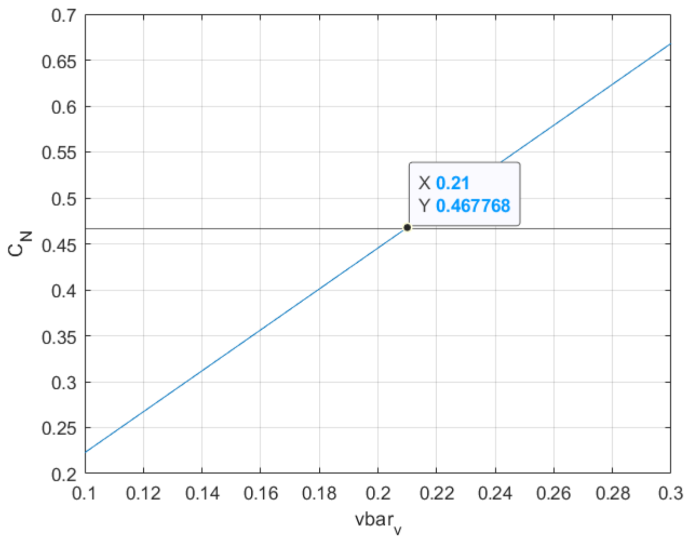

Figure 4: Yawing Moment Produced with Optimised Tail Volume Coefficient

By the inputting the optimised $\bar{V}_V$, the design process was done iteratively until all the stability conditions were met. Under maximum rudder deflection, the counter yawing moment produced is given by:

$$C_{N_{max}} = C_{N_{\delta_r}} \delta_r = 0.4678Nm \tag{36}$$

The final step was to find the hinge moment required by the rudder. Equations 27 and 37 were used.

$$C_{H_r} = C_{N_\beta} \beta + C_{N_{\delta_r}} \delta_r \tag{37}$$

which gives the hinge moment to be 1.9455 Nm.

5 Conclusion

The iterative tailplane design process was completed by establishing the design requirements at the very start and applying the methodology outlined above. The aircraft met all the stability conditions with the current tailplane and control surfaces size, align with the project objectives to keep the drone compact. In conclusion, the UAV was indeed stable.

However, the current design is far from optimal. If given more time on the project, deeper analysis could be done to justify the decisions made. For instance, analysis on coordinated turn using both ailerons and rudders that takes into account the effect of coupled states. Another point is that the elevator might be over-designed based on feedback from the Simulations Team. Implementation of an all moving tail might be better since the deflection needed by the elevator at trim is relatively small. Another alternative to investigate would be implementing canards. Different tail configurations could be analysed and compared to justify for the current design being optimal. Besides, other critical conditions could be considered such as failure of pusher propeller or one of the VTOL propellers in flight which will lead to very complex analysis.

References

[1] Wingcopter 198. https://wingcopter.com/wingcopter-198. [Accessed: 19th June 2022].

[2] DRS RQ-15 neptune. https://www.unols.org/sites/default/files/Neptune%20UAV.pdf. [Accessed: 19th June 2022].

[3] Plyrotech. Plyrotech XV-H. https://www.plyrotech.com/products/xv/. [Accessed: 19th June 2022].

[4] ElevonX. Tango - UAV VTOL. [Video]. https://www.elevonx.com/solutions/skyeye/, 2021. [Accessed: 19th June 2022].

[5] Konstantin I. Matveev. Alex E. Ockfen. International journal of naval architecture and ocean engineering. *Aerodynamic characteristics of NACA 4412 airfoil section with flap in extreme ground effect.*, 1:1–12., 2009;.

[6] J. Roskam. *Airplane Design*, chapter 1: Preliminary Sizing of Airplanes. Roskam Aviation and Engineering Corporation;, Ottawa, Kansas:, 1985.

[7] Mohammad H. Sadraey. *Aircraft Design: A Systems Engineering Approach.* John Wiley & Sons Ltd;, West Sussex:, 2013.

[8] Anthony Hasier. Fregata flying liferaft: Aerodynamics. 2022.

[9] Jordi Frittoli. Fregata flying liferaft: Propulsion and systems. 2022.

[10] Dr Errikos Levis. *Intro to Aerospace*, chapter 7: Static Stability & Trim. Imperial College London., 2019.

[11] Bojun Vito Yang. Fregata flying liferaft: Performance, stability control. 2022.

[12] Pierluigi Della Vecchia Salvatore Corcione Vincenzo Cusati. Fabrizio Nicolosi, Danilo Ciliberti. A comprehensive review of vertical tail design. 2016. [Accessed: 19th June 2022].

Appendix

The effectiveness of the control surfaces is related to the geometry of the tailplane. The relation is presented below.

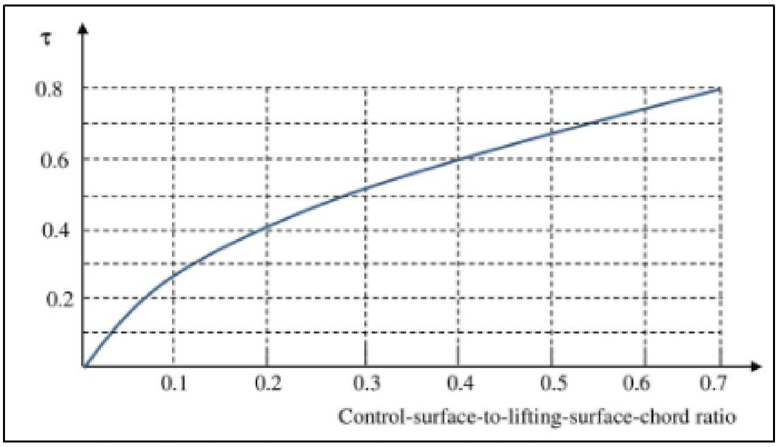

Figure 5: Effectiveness Against Chord Ratio

Empennage design and static stability analysis depends on the wing geometry and the position of centre of gravity. It was considered the middle stage of the whole project which was followed by dynamic stability analysis and testing and simulation. Table 4 shows some important parameters from Aerodynamics and Propulsion & Systems teams that were used in analysis presented in this report.

Table 4: Parameters from Other Teams

Parameter	Value
Maximum take-off weight, MTOW	51.5 kg
Centre of gravity, x_{cg}	0.6143 m
Aerodynamic centre of wing, x_{ac_w}	0.6143 m
Aspect ratio, Æ	6.83
Wing span,b	2.2 m
Wing area S_{ref}	0.7087m^2
Wing mean chord,$\bar{c}$	0.3377 m
Taper ratio, λ	0.45
Sweep,	0 °
Twist, α_{twist}	-4.5 °
Wing setting angle, i_w	1.31 °
Wing lift curve slope, a_w	4.71 rad^{-1}
Wing zero lift angle of attack, α_0	-2.71 deg
Wing stall angle, α_{stall}	14 deg
Wing maximum lift coefficient, C_{L_w}	1.26